

氢气相关化学习题册

Hydrogen-related Chemistry Exercise Book

一、选择题

I. Multiple-choice Questions

1. 点燃氢气前，一定要()。

Before igniting hydrogen, it is necessary to ().

A. 检验纯度 / Check purity

B. 通过氧化铜 / Pass through copper oxide

C. 加热 / Heat

D. 冷却 / Cool

2. 检验氢气的纯度时，集满气体的试管靠近火焰时发出爆鸣声，说明()。

When testing the purity of hydrogen, a popping sound is heard when the test tube filled with gas is close to the flame, indicating ().

A. 氢气是纯净的 / The hydrogen is pure

B. 氢气中混有氧气 / The hydrogen is mixed with oxygen

C. 氢气中混有氮气 / The hydrogen is mixed with nitrogen

D. 氢气中混有水 / The hydrogen is mixed with water

3. 用氢气还原氧化铜时，先通氢气后加热的原因是()。

When reducing copper oxide with hydrogen, the reason for passing hydrogen first and then heating is ().

A. 使氢气的浓度更大些 / To make the concentration of hydrogen higher

B. 加快反应速率 / To speed up the reaction rate

C. 使氢气先与空气中的氧气反应/ To make hydrogen react with oxygen in the air first

D. 排去试管内气/ To expel the air in the test tube

4. 下列说法中，不正确的是()。

Among the following statements, the incorrect one is () .

A. H_2 在空气中燃烧，火焰呈淡蓝色/ H_2 burns in the air with a pale blue flame

B. 利用 H_2 的还原性可以进行金属冶炼/ The reduction property of H_2 can be used for metal smelting

C. H_2 是一种新能源/ H_2 is a new energy source

D. 集满 H_2 的集气瓶应正放在桌子上/ The gas collection bottle filled with H_2 should be placed upright on the table

二、填空题

II. Fill-in-the-blank Questions

1. 氢气和氧化铜在加热条件下反应，这个反应的反应物是_____，生成物是_____。

Hydrogen and copper oxide react under heating conditions. The reactants of this reaction are____, and the products are_____.

2. 通常状况下，氢气是_____色、_____味的气体，比空气_____，_____溶于水。

Under normal circumstances, hydrogen is a_____colored,_____smelling gas,_____ than air, and_____soluble in water.

3. 实验室用排水法收集氢气的原因是_____，用向下排空气法收集氢气的原因是_____。

The reason for collecting hydrogen by the drainage method in the laboratory is_____, and the reason for collecting hydrogen by the downward air displacement method is_____.

三、实验题

III. Experiment Questions

实验室用氢气还原氧化铜，主要有以下几个操作步骤:①加热盛有氧化铜粉末的试管；②通入氢气；③ 检验氢气的纯度；④撤去酒精灯；⑤继续通入氢气，直到试管冷却。正确的操作步骤是 ()。

Hydrogen is used to reduce copper oxide in the laboratory. There are mainly the following operation steps: ① Heat the test tube containing copper oxide powder; ② Pass in hydrogen; ③ Check the purity of hydrogen; ④ Remove the alcohol lamp; ⑤ Continue to pass in hydrogen until the test tube cools down. What is the correct operation sequence? ()

- A. ③②①⑤④ B. ③②①④⑤ C. ③①②⑤④ D. ①③②④⑤

四、解答题

IV. Short Answer Questions

氢气在空气中燃烧，这个过程是物理变化还是化学变化？

When hydrogen burns in the air, is this process a physical change or a chemical change?